

MARINGA, Brazil

WMO: 868990

Lat: 23.405S

Lon: 51.933W

Elev: 1783

StdP: 13.77

Time Zone: -3.00 (SAB)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	46.4	49.9	35.2	32.1	60.0	39.0	37.3	63.4	20.2	70.9	18.6	69.3	6.9	210	0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.4	95.0	70.2	93.0	70.7	91.1	70.8	77.8	85.2	76.9	84.6	76.0	83.6	10.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	143.4	80.0	75.1	140.9	79.6	73.7	134.2	78.9	42.9	85.6	41.8	84.6	40.9	83.6	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	18.1	16.1	DB	40.8	98.5	2.5	1.7	39.0	99.8	37.6	100.8	36.2	101.8	34.4	103.1	(4)
(5)			WB	39.2	79.8	3.2	1.4	36.9	80.8	35.1	81.5	33.3	82.3	31.0	83.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.6	77.8	78.8	77.3	74.0	68.6	66.2	67.3	69.8	73.8	75.3	76.8	77.6	(6)	
	DBStd	6.84	3.19	3.63	3.52	4.06	5.38	5.82	6.72	7.71	7.50	5.67	4.19	3.59	(7)	
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)	
	HDD65	205	0	0	0	1	29	56	59	45	13	2	0	0	(9)	
	CDD50	8608	863	808	846	720	577	486	538	614	713	784	805	854	(10)	
	CDD65	3337	398	388	381	272	142	92	130	193	276	320	355	390	(11)	
	CDH74	27048	3068	3238	3057	1887	729	426	789	1684	2821	2996	3135	3218	(12)	
(13)	CDH80	9967	1087	1323	1132	559	123	52	144	611	1247	1291	1238	1160	(13)	
(14)	Wind	WSAvg	7.5	7.3	7.3	7.2	6.8	6.7	7.0	7.4	8.6	8.7	8.3	8.1	6.9	(14)
(15) Precipitation	PrecAvg	64.2	9.0	7.3	5.4	4.0	5.4	3.9	3.1	2.3	5.3	5.7	5.9	7.0	(15)	
	PrecMax	88.3	14.9	12.8	10.9	10.6	13.7	13.3	10.5	7.6	12.8	13.0	15.9	11.3	(16)	
	PrecMin	50.1	2.5	1.7	2.7	0.3	1.3	0.2	0.4	0.0	0.6	1.7	1.8	1.2	(17)	
	PrecStd	9.6	3.2	3.0	2.2	2.1	3.3	3.0	2.5	2.1	2.8	3.1	2.9	2.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	96.7	93.3	89.8	86.3	83.9	86.1	92.8	97.1	97.2	94.9	93.4	(19)	
		MCWB	73.7	70.5	72.0	71.8	70.3	67.0	64.7	65.3	67.7	70.7	73.4	71.4	(20)	
	2%	DB	90.2	93.2	91.2	87.7	82.8	81.0	83.8	89.4	94.2	94.7	91.7	91.0	(21)	
		MCWB	74.3	73.1	72.2	71.4	69.5	66.3	64.3	64.7	67.6	70.6	71.9	73.2	(22)	
	5%	DB	88.0	91.0	88.9	85.8	80.7	79.0	81.0	86.3	91.2	91.1	89.5	88.2	(23)	
		MCWB	74.3	73.6	72.3	70.3	68.0	65.9	63.5	64.3	66.2	69.5	70.1	73.2	(24)	
	10%	DB	86.0	87.9	86.3	83.9	78.5	75.7	78.9	84.0	87.6	87.5	87.1	86.1	(25)	
		MCWB	74.2	73.5	72.0	69.6	67.1	65.4	63.3	63.9	65.9	69.3	70.1	72.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.1	78.2	77.8	76.0	73.8	73.1	70.1	70.9	74.4	75.9	77.4	79.6	(27)	
		MCDB	85.9	86.7	85.0	82.1	81.0	77.3	76.6	80.3	88.1	85.0	87.2	85.1	(28)	
	2%	WB	77.7	77.0	76.3	74.6	72.3	70.4	68.3	67.6	71.8	74.0	75.9	77.4	(29)	
		MCDB	84.8	85.8	84.2	81.9	79.3	74.8	75.8	81.0	84.0	85.7	85.0	84.4	(30)	
	5%	WB	76.6	76.1	75.0	73.4	70.7	68.4	66.5	66.2	69.9	72.6	74.6	76.2	(31)	
		MCDB	83.6	84.2	83.3	80.7	77.0	74.2	74.7	79.4	81.2	84.2	83.1	83.3	(32)	
	10%	WB	75.6	75.2	73.8	72.0	69.0	66.8	65.1	64.8	68.2	71.2	73.3	75.1	(33)	
		MCDB	82.5	82.7	81.9	78.7	75.0	72.8	74.0	78.5	79.2	81.7	81.2	81.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	17.4	17.7	17.7	16.4	16.4	18.8	20.5	21.4	19.6	19.1	16.8	(35)	
		MCDBR	19.2	21.2	20.4	19.8	18.4	18.6	20.4	22.5	23.2	24.0	22.1	19.8	(36)	
	5% WB	MCWBR	8.4	7.1	7.2	8.5	8.5	8.6	8.4	8.8	8.8	9.4	8.2	8.8	(37)	
		MCWBR	16.3	17.0	17.1	16.8	15.8	15.9	18.1	20.6	20.4	19.8	17.8	16.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.395	0.391	0.388	0.368	0.351	0.368	0.368	0.428	0.532	0.455	0.405	0.399	(40)	
		taud	2.425	2.434	2.434	2.477	2.483	2.433	2.416	2.201	1.921	2.211	2.356	2.395	(41)	
	Ebn at Noon		301	298	290	282	274	261	266	259	244	276	296	300	(42)	
		Edh at Noon	40	38	37	32	30	30	31	42	60	48	42	41	(43)	
(44) All-Sky Solar Radiation	RadAvg		1879	1834	1732	1519	1182	1077	1164	1420	1575	1720	1924	1957	(44)	
	RadStd		179	150	120	105	131	110	123	98	156	143	159	189	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page